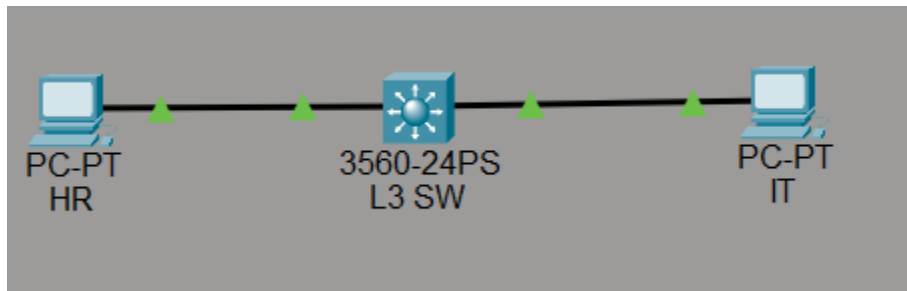


MINI PROJECT 3

Inter-VLAN Routing using Layer-3 Switch (SVI) + VLAN + Hardware Routing

DIAGRAM:



DESCRIPTION:

This project replaces the router-on-a-stick design with a Layer-3 switch to perform inter-VLAN routing using SVIs. L-3 Switch maintains Mac-address & Routing Table both routing in hardware.

STEP 1: Create VLANs on SW 1:

Verify existing VLANs from Previous projects

STEP 2: Assign Access Ports to VLANs:

```
interface fastEthernet0/1
switchport mode access
switchport access vlan 10
exit
```

```
interface fastEthernet0/2
switchport mode access
switchport access vlan 20
exit
```

STEP 3: Create SVIs (Switched Virtual Interfaces)

```
interface vlan 10
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
```

```
interface vlan 20
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
```

STEP 4: Configure IP Addressing on PCs:

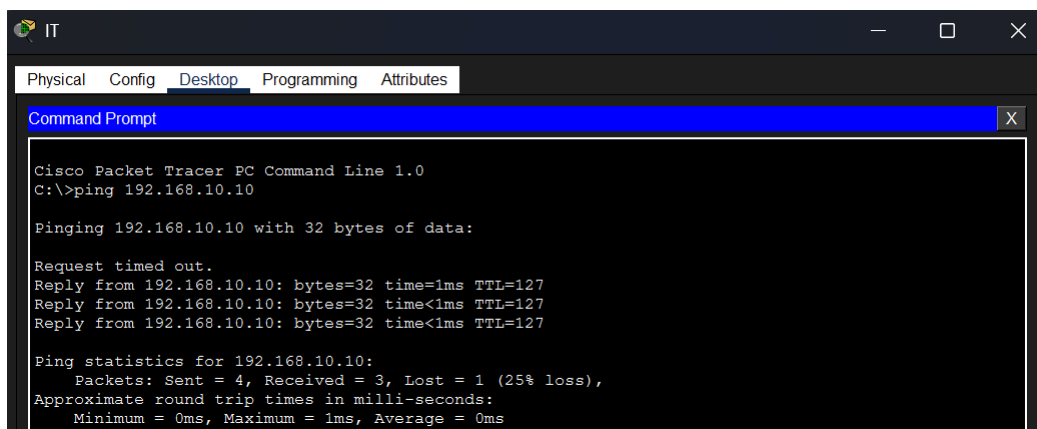
PC	VLAN	IP Address	Gateway
PC-HR	10	192.168.10.10/24	192.168.10.1
PC-IT	20	192.168.20.10/24	192.168.20.1

STEP 5: Enable IP Routing on the Switch:

```
ip routing
```

STEP 6: Verification & Ping Test:

```
show ip route
show ip interface brief
```



The screenshot shows a Cisco Packet Tracer PC Command Line window for a PC named 'IT'. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, showing a Command Prompt. The Command Prompt displays the output of a ping command: 'C:\>ping 192.168.10.10'. The output shows that the ping was successful, with 4 packets sent, 3 received, and 1 lost (25% loss). The approximate round trip times in milliseconds are: Minimum = 0ms, Maximum = 1ms, Average = 0ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.10: bytes=32 time=1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```